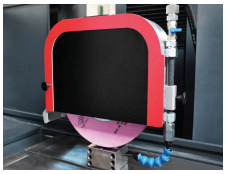
"THE FUTURE IS HERE" · SOLD & SERVICED IN CANADA BY NOVOTECH

NC SURFACE GRINDER · TGH

PRECISION SURFACE GRINDING, CROSSWISE AND LENGTHWISE









Since 1974
MANUFACTURING EXPERIENCE

17.7 × 35.4"
TABLE (450 × 900 MM)

10"
TOUCHSCREEN

Canada
PARTS & TECHNICAL SUPPORT

NovoTech Machine Tools Inc.
1365 Mid-Way Blvd, 26, Mississauga, ON L5T 2J5, Canada
NovoTechMachineTools.com | info@NovoTechMachineTools.com | +1 647 526 5510

MANUFACTURER · ESTABLISHED 1974

ABOUT DENER

Dener Makina was established in 1974 in the Kayseri Old Industrial Zone. Since 2000, the company has focused on the production of CNC Sheet Metal Processing Machines, manufacturing a wide range of products including Guillotine Shears, Hydraulic Press Brakes, Ball Screw Press Brakes, Servo-Electric Press Brakes, Plasma Cutting Machines, and Fiber Laser Cutting Machines.

In 2016, the company made a major investment by expanding its facilities to include 1,670,000 sq ft (155,000 m^2) of factory buildings and 290,000 sq ft (27,000 m^2) of social facilities on a 14,000,000 sq ft (1,300,000 m^2) site within the İncesu Organized Industrial Zone (O.S.B). In this new phase, Dener Makina began the production of CNC Metal Processing Machines such as Taksan Integrated Systems, Double Column Machining Centers, 5-Axis Simultaneous Machining Centers, Vertical Machining Centers, Lathes, and Grinding Machines.

With its extensive production range, over half a century of experience, skilled workforce, and multinational R&D center, Dener Makina holds a leading position globally by providing high-value industrial investment products in the fields of CNC Sheet Metal Processing and CNC Metal Processing Machines.

INCLUDED WITH EVERY NOVOTECH MACHINE

- ✓ Made in Turkey with European components
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ **7-day money-back guarantee** (in-stock machines only)
- ✓ Spare parts & technical support readily available in Canada
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- ✓ **Installation and training available**

SOLD & SERVICED IN CANADA

by NovoTech Machine Tools — Mississauga, ON

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Overview 04
Standard / Optional Equipment 05
Body and Column Structure 06
Working Area and Table Dimensions 07
Control and Hydraulic Unit 08
Technical Data 09
Dener Product Range 11

DENER MAKINA · İNCESU ORGANIZED INDUSTRIAL ZONE, KAYSERI

DENER PRODUCTION CAMPUS



Dener production campus, Kayseri — 1,670,000 sq ft (155,000 m^2) of factory buildings on a 14,000,000 sq ft (1,300,000 m^2) site

- SHEET PROCESSING MACHINES PRODUCTION PLANT
- MACHINE TOOLS PRODUCTION PLANT
- PRODUCTION PLANT
- SHEET WELDING FACTORY
- SHEET CLADDING FACTORY
- TECHNOLOGY CENTER
- ADMINISTRATIVE BUILDING
- SOCIAL FACILITIES
- MAIN ENTRANCE
- VISITOR PARKING AREA
- HELICOPTER PAD

52+
YEARS OF EXPERIENCE

100+
EXPORT COUNTRIES

14,000,000 sq ft
PRODUCTION CAMPUS (1,300,000 M^2)

TGH SERIES NC SURFACE GRINDING MACHINE

OVERVIEW



TGH-B 450 NC surface grinder



The wheel guard and the coolant nozzles

With the precision surface grinding machine, all kinds of materials can be ground smoothly, crosswise and lengthwise.

- Easy to use with a 10" touchscreen
- Movement on the cross and vertical axes with precision ball screws
- Precise infeed with the NC control unit on the cross and vertical axes
- Electronic-hydraulic control of the longitudinal feed
- Vibration-free, direct-drive, super-precise, powerful and rigid spindle supported at many points
- Special coating and precise scraping on the V guideways
- Precise scraping on the prismatic box guideways
- Fast service and spare parts support

TGH SERIES NC SURFACE GRINDING MACHINE

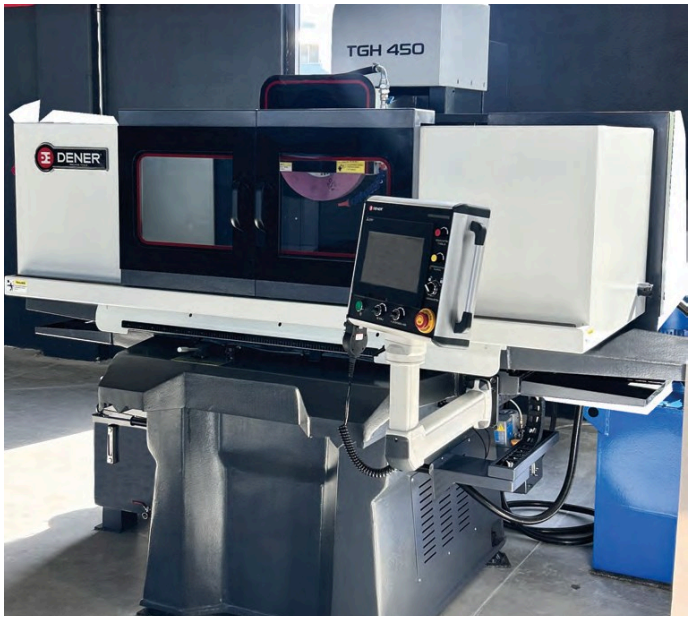
STANDARD / OPTIONAL EQUIPMENT

STANDARD EQUIPMENT

- Movement with AC servo motors on the cross and vertical axes
- 10" touchscreen
- External wired handwheel
- Table guard sheet
- Wheel flange
- Electromagnetic table (15.7 × 31.5", 400 × 800 mm)
- Wheel balancing stand
- Balancing arbor
- Grinding wheel (Ø13.8 × Ø5 × 1.57", Ø350 × Ø127 × 40 mm)
- Wheel flange puller
- Wheel dressing device on the table (diamond)
- Coolant equipment
- Paper-filter coolant equipment (manual)
- Alarm / warning and lighting lamps
- Front door safety switch
- Tools and tool kit
- Hydraulic unit and hydraulic oil (26 US gal, 100 L)
- Demagnetizing unit for the electromagnetic table
- Lifting bolts
- Levelling bolts and plates
- User manual
- Mitsubishi NC control on the cross and vertical axes

OPTIONAL EQUIPMENT

- Additional wheel flange
- Wheel dressing device on the wheel head (diamond)
- Magnetic sine plate (V-S 100M)
- Magnetic and paper-filter coolant equipment
- Precision toolmaker's vise (VMV-40)
- Demagnetizer for the workpiece (VDM-68)
- Sine bar (VS-100)
- Profile grinding and dressing device (V-PB-BS)





TGH SERIES NC SURFACE GRINDING MACHINE

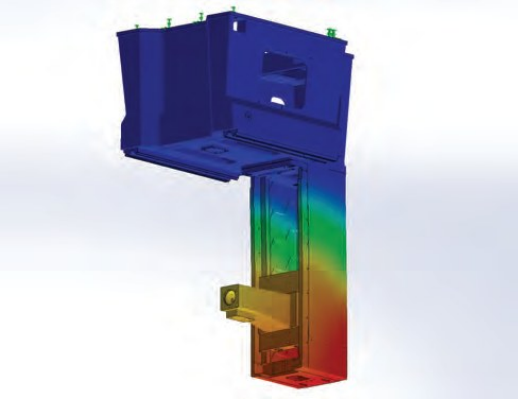
BODY AND COLUMN STRUCTURE

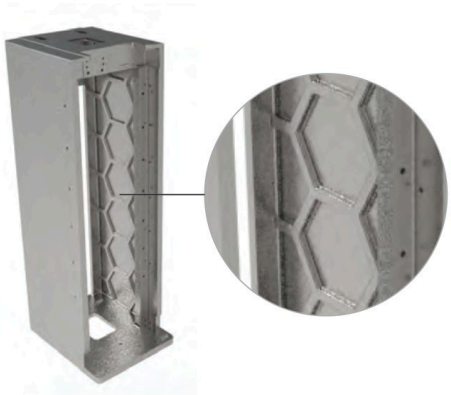
BODY STRUCTURE

Stress-relieved cast body structure for working in heavy conditions, vibration damping, high precision and high durability.









FINITE ELEMENT METHOD

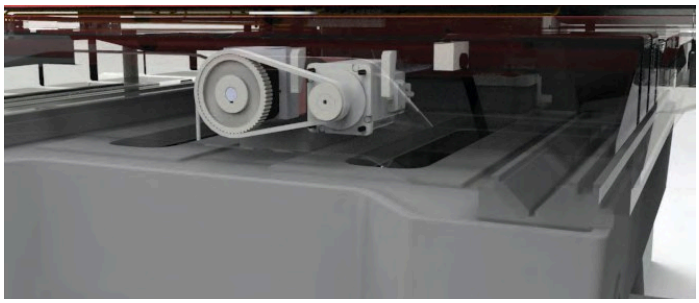
The optimum body structure obtained using the finite element method.

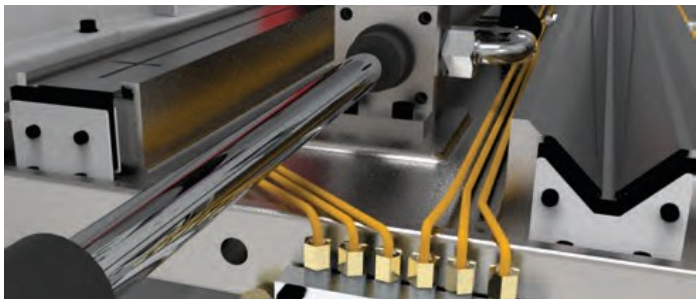
COLUMN STRUCTURE

High durability and vibration damping thanks to the honeycomb ribs inside the column.

BEARING SYSTEM

Precise support, positioning and vibration-free operation with the Turcite-B coated box-prismatic guideway system machined and scraped on the cast body.





TGH SERIES NC SURFACE GRINDING MACHINE

WORKING AREA AND TABLE DIMENSIONS

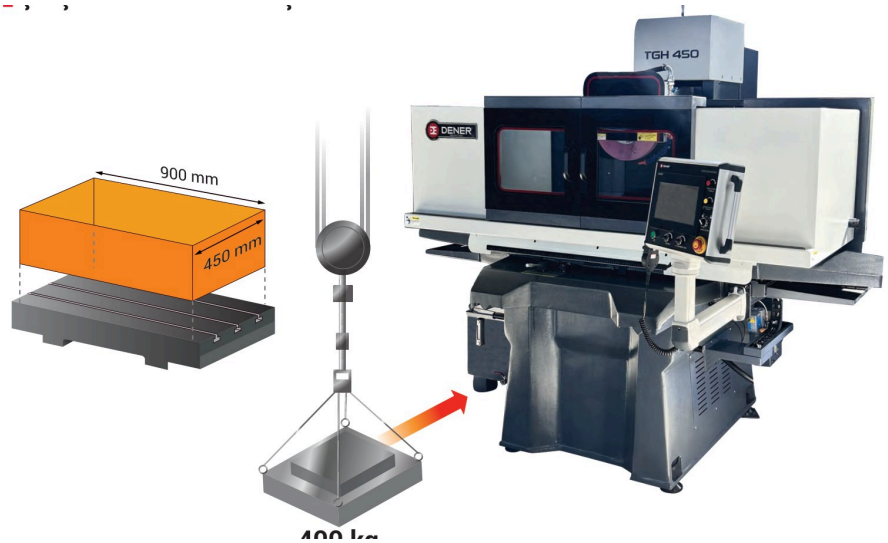
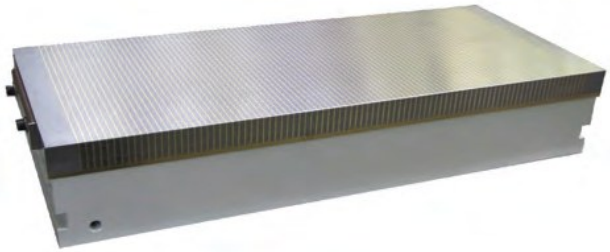


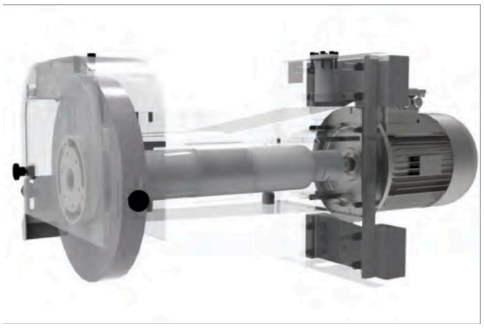
Table 35.4 × 17.7" (900 × 450 mm); parts up to 882 lb (400 kg) loaded by crane

MAGNETIC TABLE



Electromagnetic table 15.7 × 31.5" (400 × 800 mm)





COLUMN AXIS

Rigid support on hardened, Turcite-B coated rails, and precise positioning and repeatability with a high-precision, preloaded, ground ball screw.

DIRECT-DRIVE SPINDLE

An excellent surface and precise machining with the spindle supported on preloaded precision bearings and the direct-drive motor connection.

TGH SERIES NC SURFACE GRINDING MACHINE

CONTROL AND HYDRAULIC UNIT

CONTROL UNIT

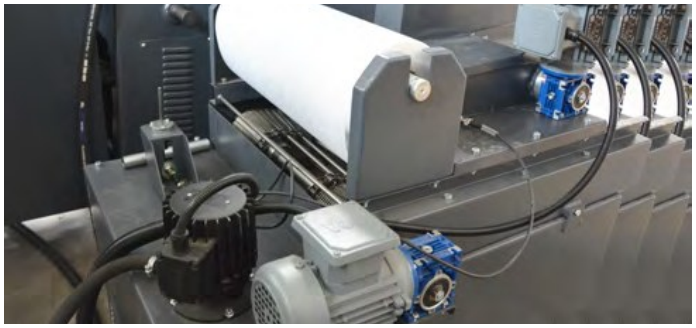
Easy to use, with up-to-date technological modules, an operator-friendly interface and an ergonomic design.





COOLANT TANK (OPTION)

Optimum cooling and easy maintenance with a 26 US gal (100 L) tank with paper and magnetic filters and electronic level control.



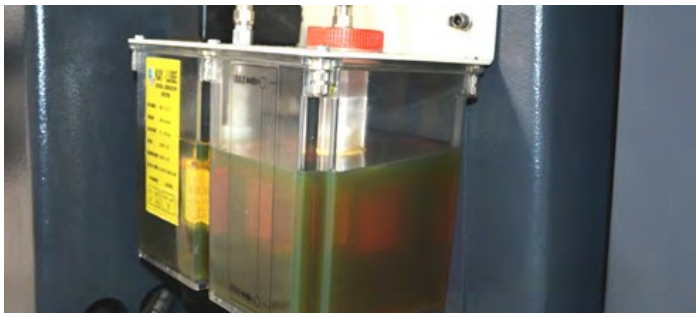
HYDRAULIC UNIT

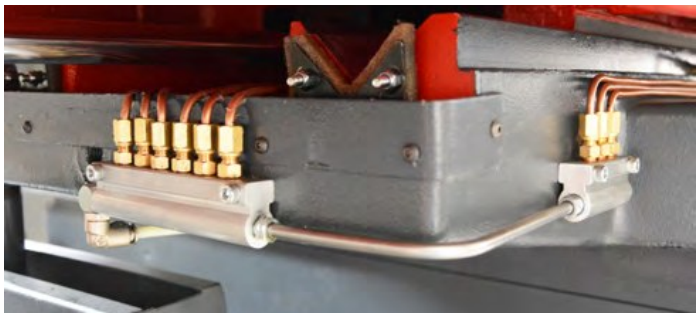
Fast, stable operation even in difficult conditions, with the electronically controlled hydraulic unit driving the table axis.



AUTOMATIC CENTRAL LUBRICATION SYSTEM

Efficient lubrication of the table axis and all the other axes with two separate pumps, extending the life of the equipment.





TGH SERIES NC SURFACE GRINDING MACHINE

TECHNICAL DATA · 1 OF 2

Model	TGH-B 450 (M)
WORK TABLE	
Table size	17.7 × 35.4″ (450 × 900 mm)
Max. grinding length (longitudinal)	35.4″ (900 mm)
Max. grinding width (cross)	17.7″ (450 mm)
Max. distance from the spindle centre to the table surface	21.7″ (550 mm)
Min. distance from the spindle centre to the table surface	3″ (75 mm)
Table guideways	V & flat way
T-slots — number / size	1 / 14H7
Max. table load	882 lb (400 kg)
BALL SCREWS	
Ball screw for the vertical axis	C3, Ø0.98″ (C3 25 mm)
Ball screw for the cross axis	C6, Ø0.98″ (C6 25 mm)
AXIS MOVEMENTS	
Rapid traverse up / down	236 in/min (6,000 mm/min)
Rapid traverse cross	236 in/min (6,000 mm/min)
Min. feed setting	0.00004″ (0.001 mm)
Table speed, approx. — longitudinal	16.4 – 82 ft/min (5 – 25 m/min)
Automatic feed — cross	0.002 – 0.79″ (0.05 – 20 mm)
Automatic feed — vertical	0.00004″ (0.001 mm)
MOTORS	
AC spindle motor power	5.4 HP (4 kW)
AC spindle motor speed	1,450 rpm
Servo motor power — vertical movement	0.9 HP (0.7 kW)
Servo motor speed — vertical movement	3,000 rpm
Servo motor torque — vertical movement	1.8 lbf·ft (2.4 N·m)
Servo motor power — cross movement	0.9 HP (0.7 kW)
Servo motor speed — cross movement	3,000 rpm
Servo motor torque — cross movement	1.8 lbf·ft (2.4 N·m)
AC motor power — longitudinal movement	5.4 HP (4 kW)
AC motor speed — longitudinal movement	1,440 rpm
AC lubrication pump motor power	0.064 HP (0.048 kW)

TGH SERIES NC SURFACE GRINDING MACHINE

TECHNICAL DATA · 2 OF 2

Model	TGH-B 450 (M)
GRINDING WHEEL	
Frequency converter type	FR-F840
Speed	100 – 1,450 rpm
Max. speed	1,450 rpm
Dimensions (outer Ø / inner Ø / thickness)	Ø13.8 × Ø5 × 1.57" (Ø350 × Ø127 × 40 mm)
POWER CONNECTION	
Electrical power connection	22.3 kVA
MACHINE DIMENSIONS	
Length	129.1" (3,280 mm)
Width	108.3" (2,750 mm)
Height	84.7" (2,151 mm)
Weight	8,157 lb (3,700 kg)
COOLANT — CAPACITY, MOTOR, PRESSURE	
Coolant pump power (pressure)	0.27 HP (5.8 psi) (0.2 kW (0.4 bar))
Coolant tank capacity	26 US gal (100 L)
HYDRAULICS — CAPACITY, MOTOR, PRESSURE	
Hydraulic pump power (pressure)	5.4 HP (580 psi) (4 kW (40 bar))
Hydraulic tank capacity	26 US gal (100 L)
CONTROL UNIT	
NC system	Mitsubishi FX5U
Screen	10" colour LCD touch screen

The values in the table may change with renewals and revisions of the machines. Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase. CSA-approved electrical panel.

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DENER PRODUCT RANGE

PRESS BRAKES


Hydraulic press brake


Tandem press brake


Servo-electric press brake


Ball-screw press brake

PANEL BENDER


Panel bender

SHEARS


CNC hydraulic shear


NC hydraulic shear

LASER CUTTING MACHINES


GFL fiber laser cutting machine


Smart fiber laser cutting machine


Tube profile laser cutting machine

MACHINE TOOLS


CNC lathe


CNC vertical milling machine


Double-column milling machine


5-axis machining center


NC grinding machine

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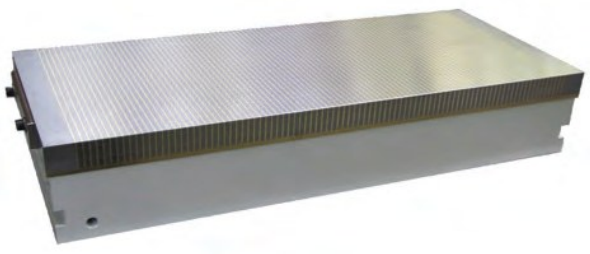
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Specifications subject to change without notice. Imperial values are converted from the manufacturer's metric data and rounded: plate thicknesses in common fractions (1/2, 1/4, 1/8, 1/16). Model numbers are manufacturer designations. Photos may show optional equipment.